

# SAT UZBEKISTAN

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A research manager selected 2 random samples of ovens of a certain type to estimate the average amount of time this type of oven takes to preheat to 325 degrees Fahrenheit ( $^{\circ}\text{F}$ ). The research manager recorded the amount of time, in minutes, each oven takes to preheat to 325 $^{\circ}\text{F}$ . Based on the first sample, the research manager estimated that this type of oven takes an average of 15.2 minutes to preheat to 325 $^{\circ}\text{F}$ , with an associated margin of error of 1 minute. Based on the second sample, the research manager estimated that this type of oven takes an average of 15.4 minutes to preheat to 325 $^{\circ}\text{F}$ , with an associated margin of error of 2.2 minutes. Assuming the margins of error were calculated the same way, which of the following best explains why the first sample obtained a smaller margin of error than the second sample?

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Mark for Review

- (A) The first sample contained more ovens than the second sample.
- (B) The first sample contained fewer ovens than the second sample.
- (C) The first sample took more time on average to preheat to 325 $^{\circ}\text{F}$  than the second sample.
- (D) The first sample took less time on average to preheat to 325 $^{\circ}\text{F}$  than the second sample.

The equation  $N(m) = 65(Q)^{\frac{m}{120}}$  gives the predicted population  $N(m)$ , in thousands, of a certain bacteria colony  $m$  minutes after the initial measurement, where  $Q$  is a constant greater than 1. The predicted population increases by  $p\%$  every 120 seconds. What is the value of  $p$  in terms of  $Q$ ?

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Mark for Review

- (A)  $100(Q^{\frac{3}{2}} + 1)$
- (B)  $100(Q^{30} + 1)$
- (C)  $100(Q^{\frac{1}{2}} - 1)$
- (D)  $100(Q^{30} - 1)$

Ari examined a set of 86 plants. The lightest plant had a mass of 2.8 kilograms (kg) and the heaviest plant had a mass of 6.2 kg. Chihiro examined the same set of 86 plants and also an additional plant with a mass of 11.7 kg. Which of the following must be true about the set of plants that Ari examined and the set of plants that Chihiro examined?

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Mark for Review

- (A) The standard deviation of the masses, in kg, of the plants that Ari examined is greater than the standard deviation of the masses, in kg, of the plants that Chihiro examined.
- (B) The range of the masses, in kg, of the plants that Ari examined is greater than the range of the masses, in kg, of the plants that Chihiro examined.
- (C) The median of the masses, in kg, of the plants that Ari examined is less than the median of the masses, in kg, of the plants that Chihiro examined.
- (D) The mean of the masses, in kg, of the plants that Ari examined is less than the mean of the masses, in kg, of the plants that Chihiro examined.

A certain investment account offers a special interest rate for the first 4 months the account is open followed by a lower interest rate for the remainder of the time the account is open. Akeyla opened one of these accounts with an original account balance of \$700.00 and did not make any other deposits or withdrawals. 4 months after Akeyla opened the account, the balance had increased by 0.6% of the original balance. 6 months after Akeyla opened the account, the balance had increased by an additional 0.1% of the balance at the end of the first 4 months. Every 2 months after the first 6 months, the balance had increased by an additional 0.1% of the balance 2 months before. Which of the following equations could represent the account balance  $B(x)$ , in dollars,  $x$  months after the account was opened, where  $x \geq 4$ ?

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(A)

$$B(x) = 704.20(1.001)^{\frac{x-4}{2}}$$

(B)

$$B(x) = 700.00(1.001)^{\frac{x-4}{2}}$$

(C)

$$B(x) = 704.20(1.001)^{2(x-4)}$$

(D)

$$B(x) = 700.00(1.001)^{2(x-4)}$$

For a study, a group of chipmunks will be selected from a habitat consisting of 220 chipmunks, and a group of prairie dogs will be selected from a habitat consisting of 190 prairie dogs. Some of the chipmunks and prairie dogs will be in a treatment group, and some of the chipmunks and prairie dogs will be in a control group. Which of the following is necessary for this study to attempt to establish a cause-and-effect relationship between two variables?

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Mark for Review

(A)

The number of chipmunks in the treatment group is equal to the number of prairie dogs in the treatment group, and the number of chipmunks in the control group is equal to the number of prairie dogs in the control group.

(B)

The chipmunks and the prairie dogs are randomly assigned to the treatment and control groups.

(C)

The chipmunks and the prairie dogs are randomly selected from their respective habitats.

(D)

The average age of the chipmunks in the treatment group is equal to the average age of the prairie dogs in the treatment group, and the average age of the chipmunks in the control group is equal to the average age of the prairie dogs in the control group.

As the size of a firework's shell increases, the height above the ground when the firework bursts increases. For a firework to burst 1,075 feet above the ground, a firework's shell in the shape of a sphere with a volume of 542.68 cubic inches could be used. To the nearest whole number, what is the volume of this firework's shell in cubic centimeters? (1 inch= 2.54 centimeters)

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Mark for Review

$$34z^{18} + bz^9 + 30$$

In the given expression,  $b$  is a positive integer. If  $qz^9 + r$  is a factor of the expression, where  $q$  and  $r$  are positive integers, what is the greatest possible value of  $b$ ?

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Mark for Review

A right rectangular prism has a base area of  $16t$  square centimeters ( $\text{cm}^2$ ). The length of the base of the rectangular prism is  $\frac{8}{3}$  cm, and the height of the rectangular prism is 9 cm. Which expression represents the surface area, in  $\text{cm}^2$ , of the right rectangular prism?

8

Mark for Review

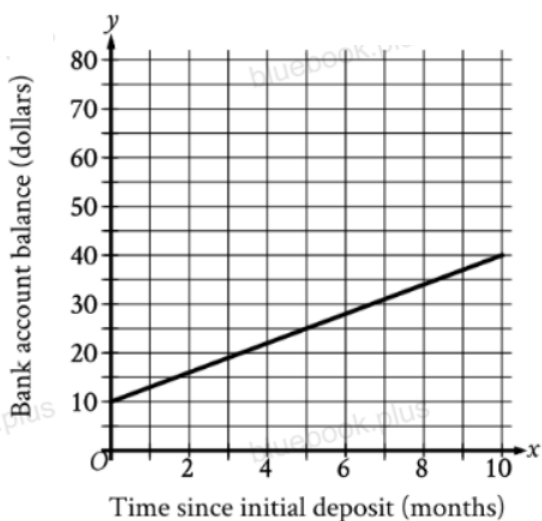
ABC

(A)  $32t + 96$

(B)  $140t + 48$

(C)  $800t + 48$

(D)  $144t$

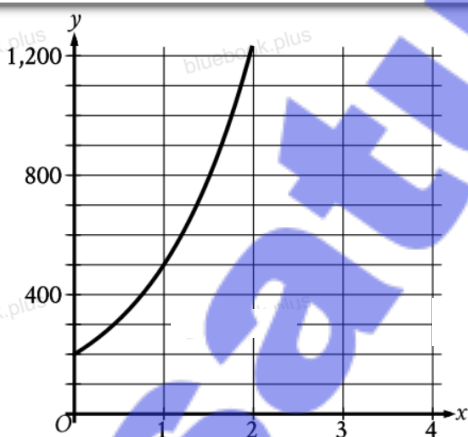


A bank account was opened with an initial deposit. Over the next several months, regular deposits were made into this account, and there were no withdrawals made during this time. The graph of the function  $f$  shown, where  $y = f(x)$ , estimates the account balance, in dollars, in this bank account  $x$  months since the initial deposit. To the nearest whole dollar, what is the amount of the initial deposit estimated by the graph?

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Mark for Review

ABC



The graph models the number of online newsletter subscribers at the end of every six-month period, where  $x$  is the number of six-month periods since the end of January 1995 and  $0 \leq x \leq 4$ . Which statement is the best interpretation of the point  $(1, 500)$  in this context?

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Mark for Review

ABC

(A) The estimated number of online newsletter subscribers at the end of the first six-month period was 500.

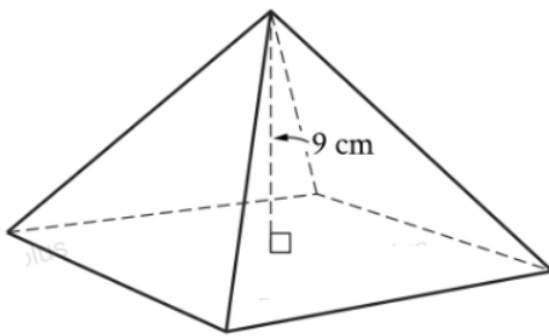
(B) The estimated number of online newsletter subscribers increased every six months by 500 subscribers.

(C) The estimated number of online newsletter subscribers at the end of January 1995 was 500.

(D) The estimated number of online newsletter subscribers at the end of January was.

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Note: Figure not drawn to scale.

The right square pyramid shown has a height of 9 centimeters (cm). The length, in cm, of one edge of the pyramid's base is  $\frac{7}{3}$  times the height of the pyramid. What is the volume, in  $\text{cm}^3$ , of the pyramid?

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Mark for Review

ABC

In 2018, approximately 20,000 kilowatts of solar power capacity was newly installed in a certain area in the United States. In 2019, the amount of newly installed solar power capacity in this area increased by 13% from the amount of newly installed solar power capacity in 2018. If the amount of newly installed solar power capacity continues to increase each year by 13% of the previous year's newly installed solar power capacity, which of the following types of functions best describes how the amount of newly installed solar power capacity changes over time?

- (A) Decreasing exponential
- (B) Increasing exponential
- (C) Decreasing linear
- (D) Increasing linear

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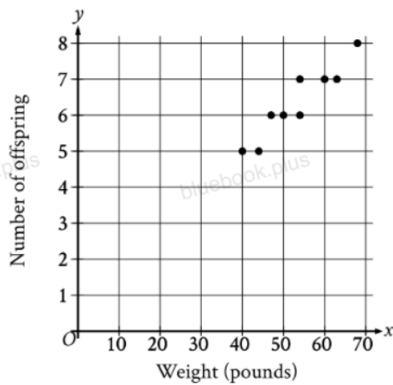
Mark for Review

ABC

An object is launched into the air from ground level. According to a quadratic model, 1.3 seconds after the object is launched, it reaches its maximum height of 27.04 feet above ground level. Which equation represents this model, where  $f(t)$  is the object's height, in feet, above ground level  $t$  seconds after it was launched?

- (A)  $f(t) = -16(t - 1.3)^2 + 27.04$
- (B)  $f(t) = 16(t - 1.3)^2 + 27.04$
- (C)  $f(t) = -16(t + 1.3)^2 + 27.04$
- (D)  $f(t) = 16(t + 1.3)^2 + 27.04$

The scatterplot shows the relationship between the weight, in pounds, of each of 9 pregnant beavers on April 30 and the number of offspring each beaver produced.



How many offspring did the 40-pound beaver produce?

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Mark for Review

(A) 5

(B) 6

(C) 7

(D) 8

A space probe uses a square-shaped solar sail to move through space. The side length of the solar sail is  $8.75w$  meters, where  $w$  is the width, in meters, of the space probe. Which equation gives the area  $A$ , in square meters, of the solar sail?

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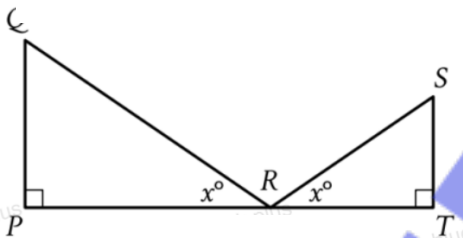
Mark for Review

(A)  $A = (w + 8.75)(w + 8.75)$

(B)  $A = (4)(8.75w)$

(C)  $A = (8.75w)(8.75w)$

(D)  $A = \left(\frac{8.75}{w}\right)\left(\frac{8.75}{w}\right)$



$\triangle QPR$  is similar to  $\triangle STR$ . The lengths represented by  $\overline{ST}$ ,  $\overline{QP}$ ,  $\overline{PR}$ , and  $\overline{QR}$  in the figure are 11, 18, 24, and 30, respectively. What is the length of  $\overline{SR}$ ?

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Mark for Review

(A)  $\frac{330}{18}$

(B)  $\frac{330}{24}$

(C)  $\frac{198}{24}$

(D)  $\frac{198}{30}$

A moving truck rental company charges its customers \$195 for the first hour of renting a truck and \$75 for each additional hour, plus the cost of gas used. Which equation represents this situation, where  $y$  is the total cost, in dollars, of renting a truck from this company for  $x$  hours,  $w$  is the cost, in dollars, of gas used, and  $x \geq 1$ ?

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Mark for Review

(A)  $y = 195 + 75(x - 1) + w$

(B)  $y = 195 + 75x + w$

(C)  $y = 195 + 75 + w$

(D)  $y = 75(x - 1) + w$

$$f(x) = (x - a)(x - b)$$

The function  $f$  is defined by the given equation, where  $a$  and  $b$  are integer constants. If  $f(15) > 0$ ,  $f(18) < 0$ , and  $f(21) > 0$ , which of the following could be the value of  $a + b$ ?

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Mark for Review

(A) 18

(B) 20

(C) 33

(D) 35

$$g(x) = \frac{1}{11} (5(2)^x + 4)$$

The function  $g$  is defined by the given equation. For all values of  $x$ , the value of  $g(x)$  is greater than  $k$ , where  $k$  is a constant. What is the greatest possible value of  $k$ ?

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Mark for Review

(A) 0

(B)  $\frac{4}{11}$

(C)  $\frac{9}{11}$

(D) 4

For a certain type of rope, the equation  $y = 900ax^2$ , where  $a$  is a constant, gives the estimated breaking strength  $y$ , in pounds, of a rope with a circumference of  $x$  inches. Based on this equation, if a rope of this type has a circumference of 2.75 inches, it has an estimated breaking strength of 9,528.75 pounds. What is the estimated breaking strength, in pounds, of a rope of this type that has a circumference of 5.50 inches?

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